

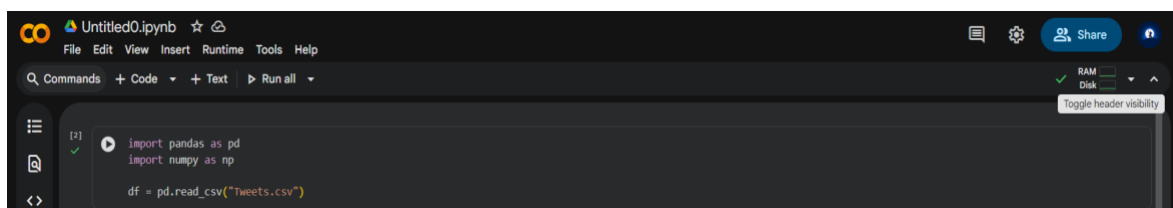
# Assignment 2

Name : Chandrashekhar Mallikarjun Kore

PRN : 202501090022

Course : Essential of data Science

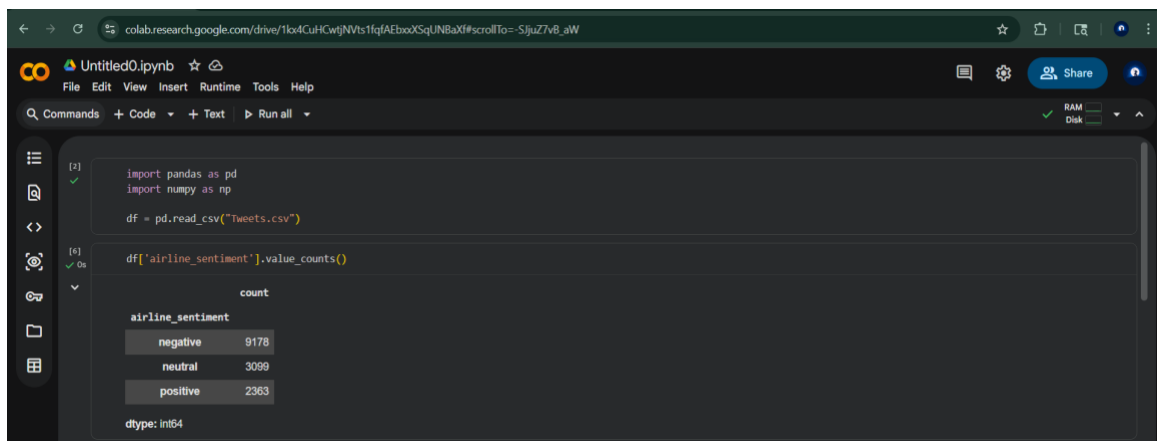
Faculty : Mrs. Supriya Sabale



```
[1]: import pandas as pd
import numpy as np

df = pd.read_csv("Tweets.csv")
```

**1.Problem:** Count total tweets for each airline\_sentiment.

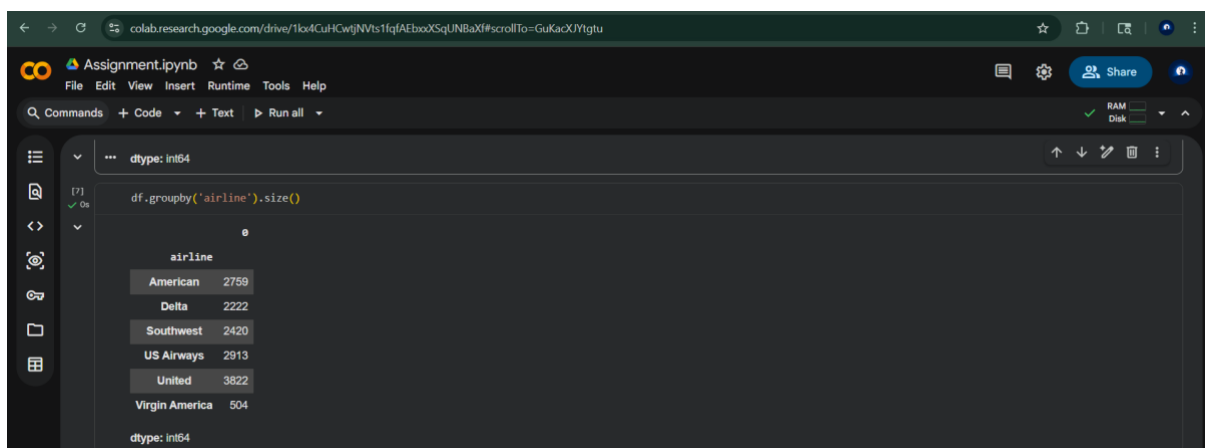


```
[6]: df['airline_sentiment'].value_counts()
```

| airline_sentiment | count |
|-------------------|-------|
| negative          | 9178  |
| neutral           | 3099  |
| positive          | 2363  |

dtype: int64

**2.Problem:** Find total tweets for each airline.

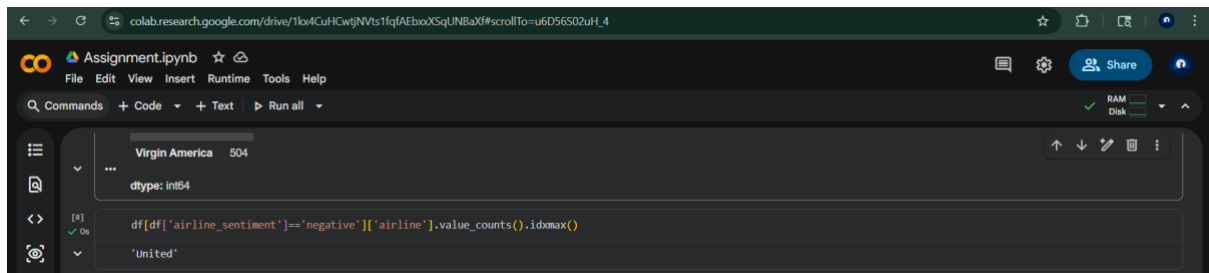


```
[7]: df.groupby('airline').size()
```

| airline        | 0    |
|----------------|------|
| American       | 2758 |
| Delta          | 2222 |
| Southwest      | 2420 |
| US Airways     | 2913 |
| United         | 3822 |
| Virgin America | 504  |

dtype: int64

### 3. Problem: Identify airline with highest number of negative sentiments.



A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell shows the following Python code:

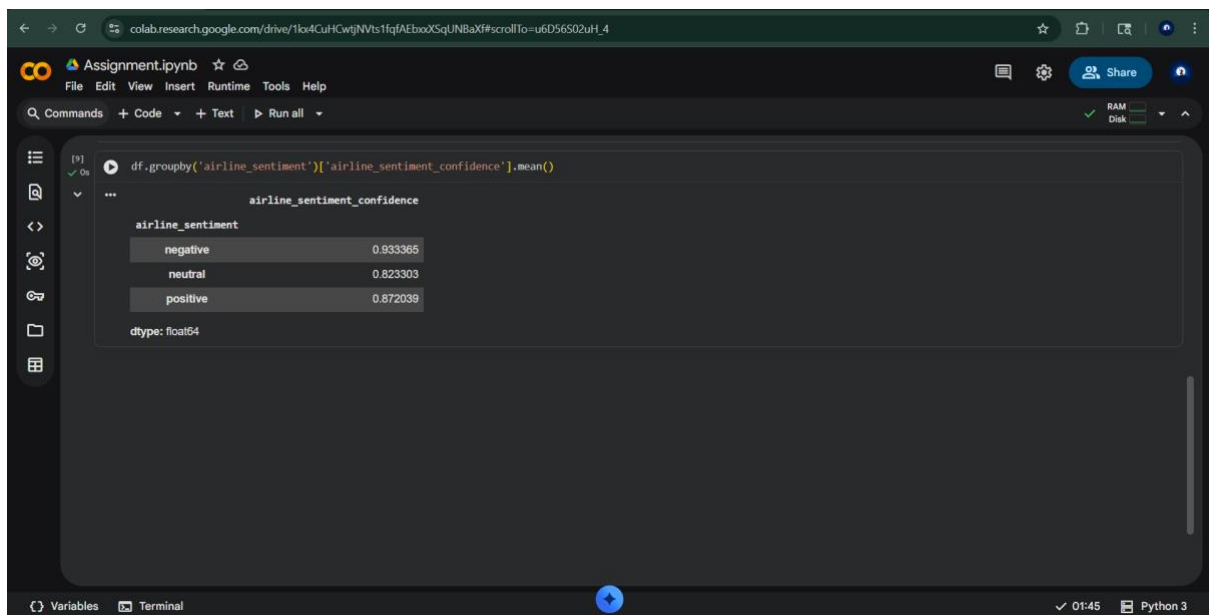
```
df[df['airline_sentiment']=='negative']['airline'].value_counts().idxmax()
```

The output of the code is displayed as a text box containing:

```
Virgin America    504
dtype: int64
```

The interface includes a left sidebar with file and variable views, a top menu bar, and a bottom status bar showing "Python 3".

### 4. Problem: Compute average airline\_sentiment\_confidence per sentiment.



A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell shows the following Python code:

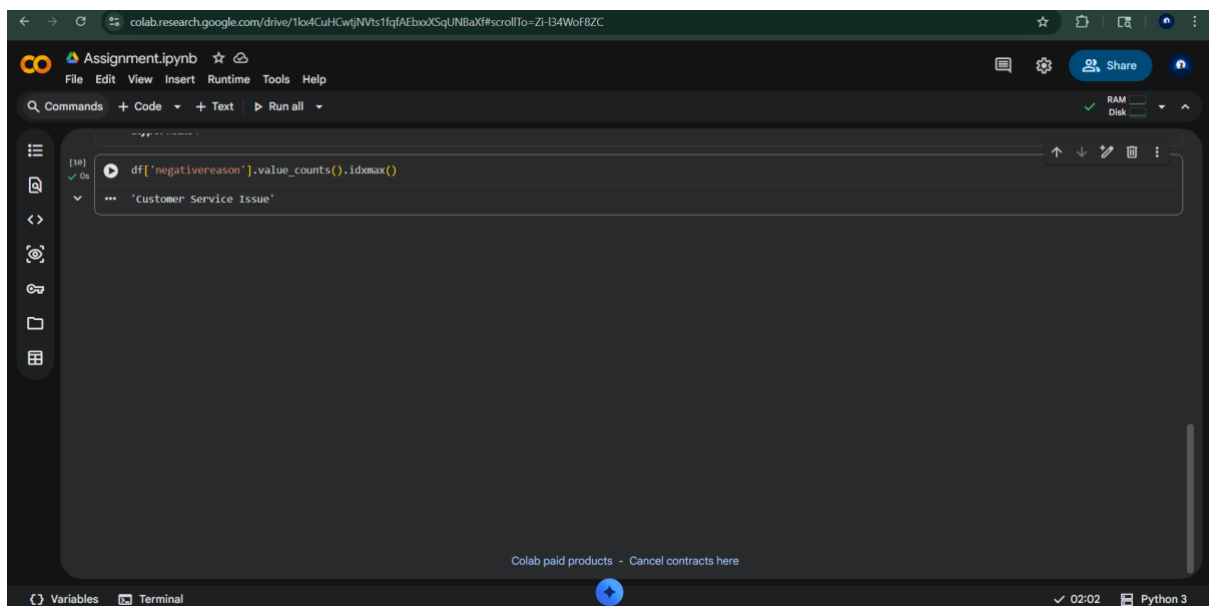
```
df.groupby('airline_sentiment')['airline_sentiment_confidence'].mean()
```

The output is a table with the following data:

| airline_sentiment | airline_sentiment_confidence |
|-------------------|------------------------------|
| negative          | 0.933365                     |
| neutral           | 0.823303                     |
| positive          | 0.872039                     |

The output is also labeled with `dtype: float64`. The interface includes a left sidebar, a top menu bar, and a bottom status bar showing "Python 3".

### 5. Problem: Find most frequent negativereason.



A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell shows the following Python code:

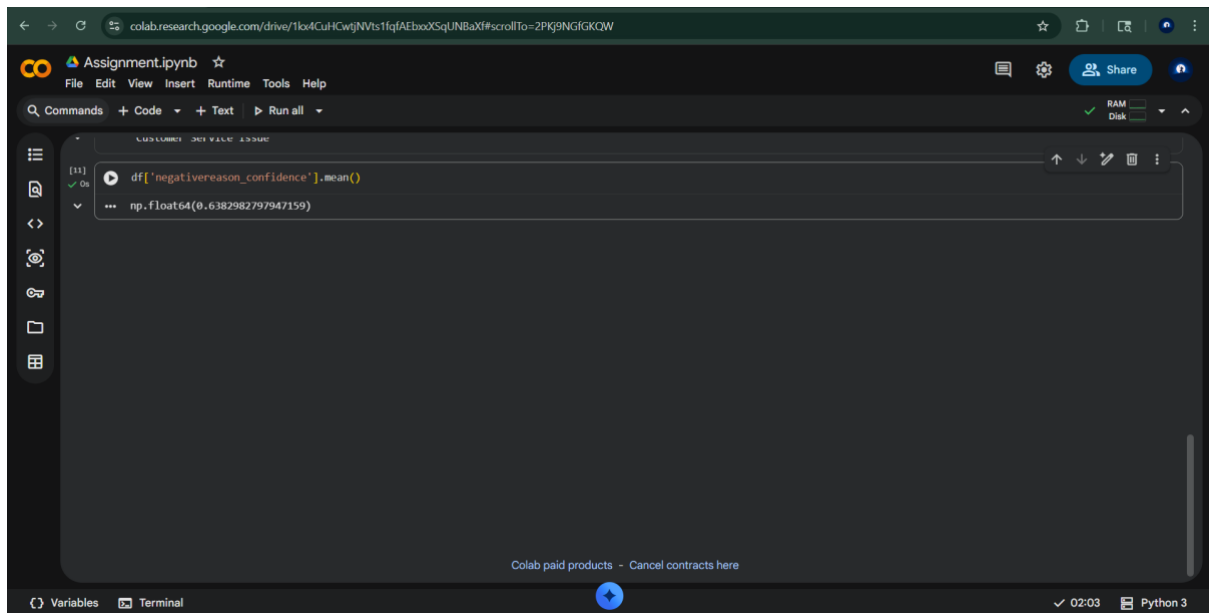
```
df['negativereason'].value_counts().idxmax()
```

The output of the code is displayed as a text box containing:

```
'Customer Service Issue'
```

The interface includes a left sidebar, a top menu bar, and a bottom status bar showing "Python 3".

## 6. Problem: Compute mean negativereason\_confidence.

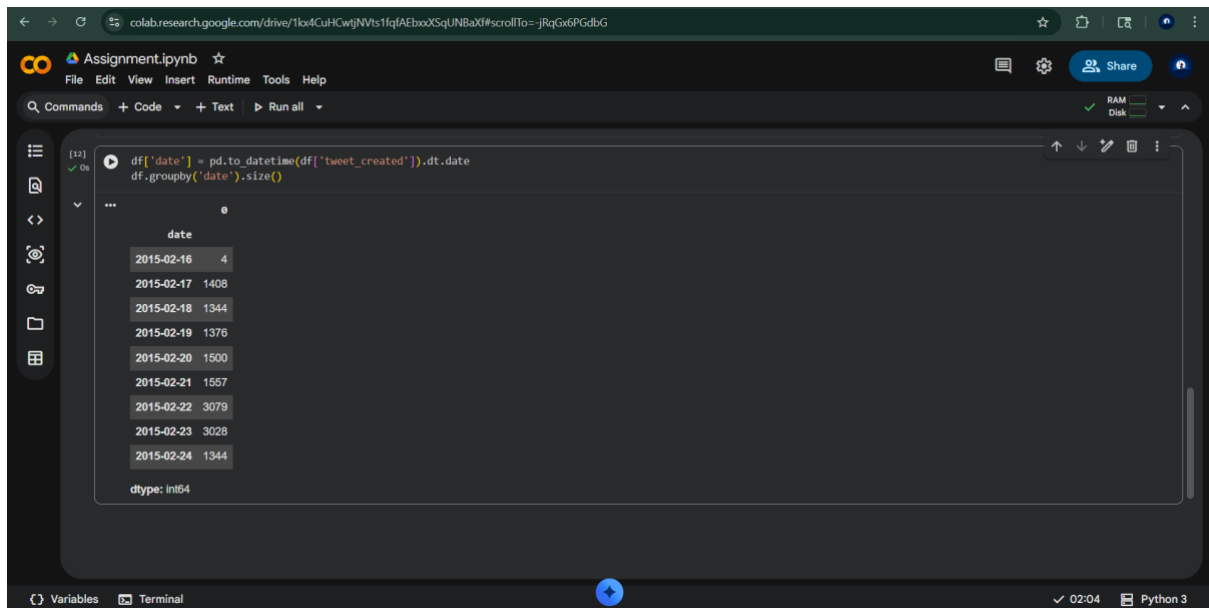


A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell [11] contains the following Python code:

```
df['negativereason_confidence'].mean()  
...  
np.float64(0.6382982797947159)
```

The output of the code is displayed below the code cell. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Variables", "Terminal", and "Python 3".

## 7. Problem: Count tweets using tweet\_created.

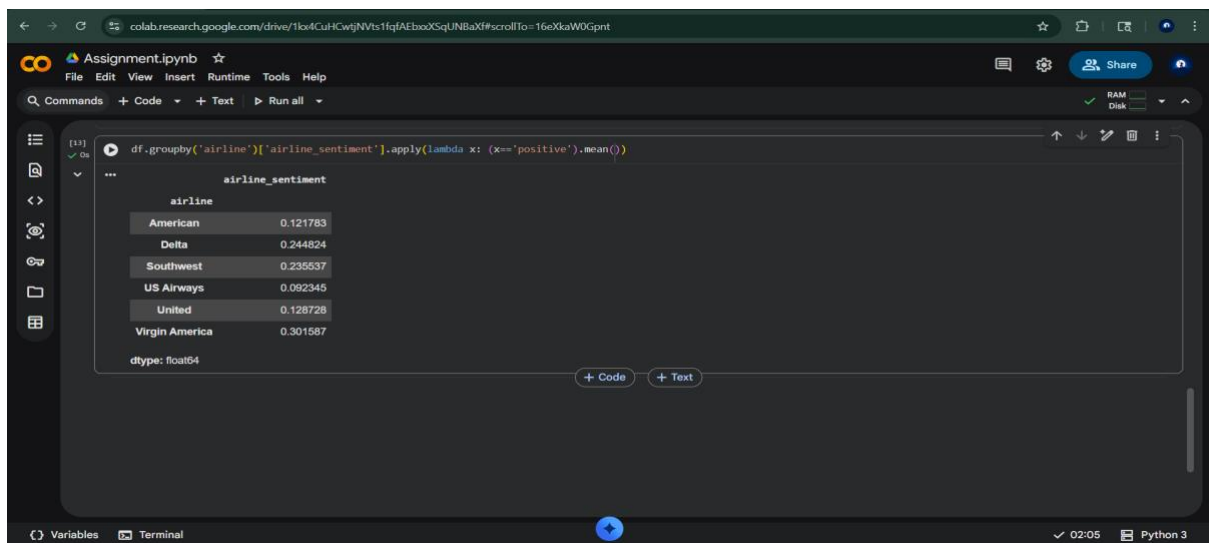


A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell [12] contains the following Python code:

```
df['date'] = pd.to_datetime(df['tweet_created']).dt.date  
df.groupby('date').size()  
...  
date  
2015-02-16    4  
2015-02-17   1408  
2015-02-18   1344  
2015-02-19   1376  
2015-02-20   1500  
2015-02-21   1557  
2015-02-22   3079  
2015-02-23   3028  
2015-02-24   1344  
dtype: int64
```

The output of the code is displayed below the code cell. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Variables", "Terminal", and "Python 3".

8. **Problem:** Find airline with highest proportion of positive tweets.



The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the following Python code:

```
df.groupby('airline')['airline_sentiment'].apply(lambda x: (x=='positive').mean())
```

The output is a DataFrame with the following data:

| airline        | airline_sentiment |
|----------------|-------------------|
| American       | 0.121783          |
| Delta          | 0.244824          |
| Southwest      | 0.235537          |
| US Airways     | 0.092345          |
| United         | 0.128728          |
| Virgin America | 0.301587          |

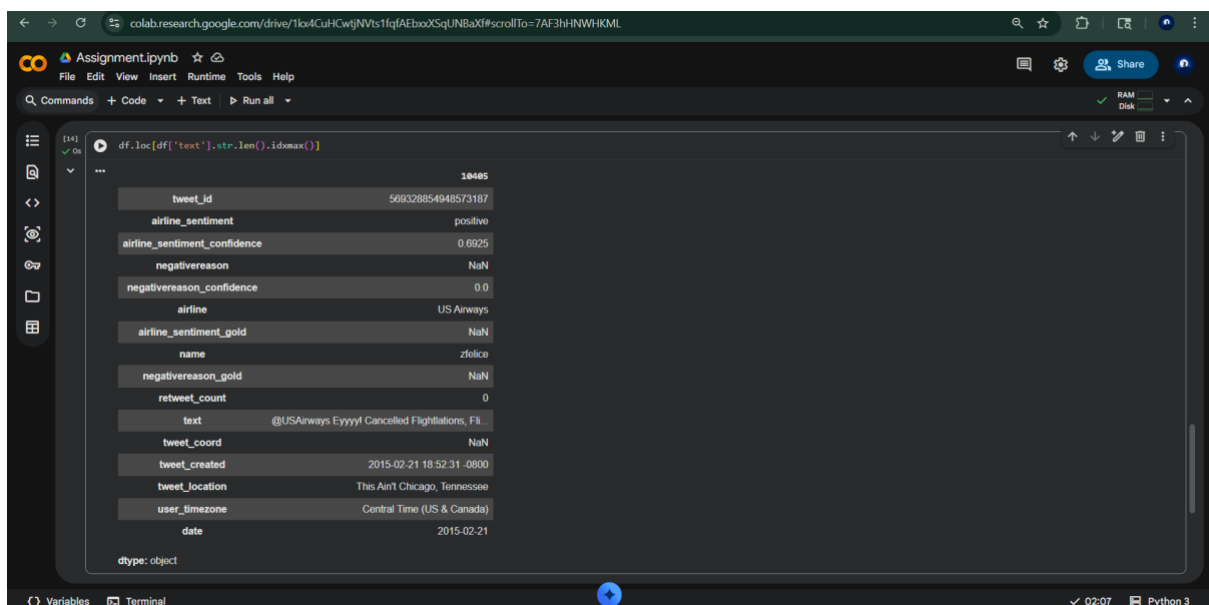
The output is a DataFrame with the following data:

| airline        | airline_sentiment |
|----------------|-------------------|
| American       | 0.121783          |
| Delta          | 0.244824          |
| Southwest      | 0.235537          |
| US Airways     | 0.092345          |
| United         | 0.128728          |
| Virgin America | 0.301587          |

The output is a DataFrame with the following data:

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|----------------|-------------------|
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| Southwest      | 0.235537          |
| US Airways     | 0.092345          |
| United         | 0.128728          |
| Virgin America | 0.301587          |

9. **Problem:** Identify tweet with maximum length from text.



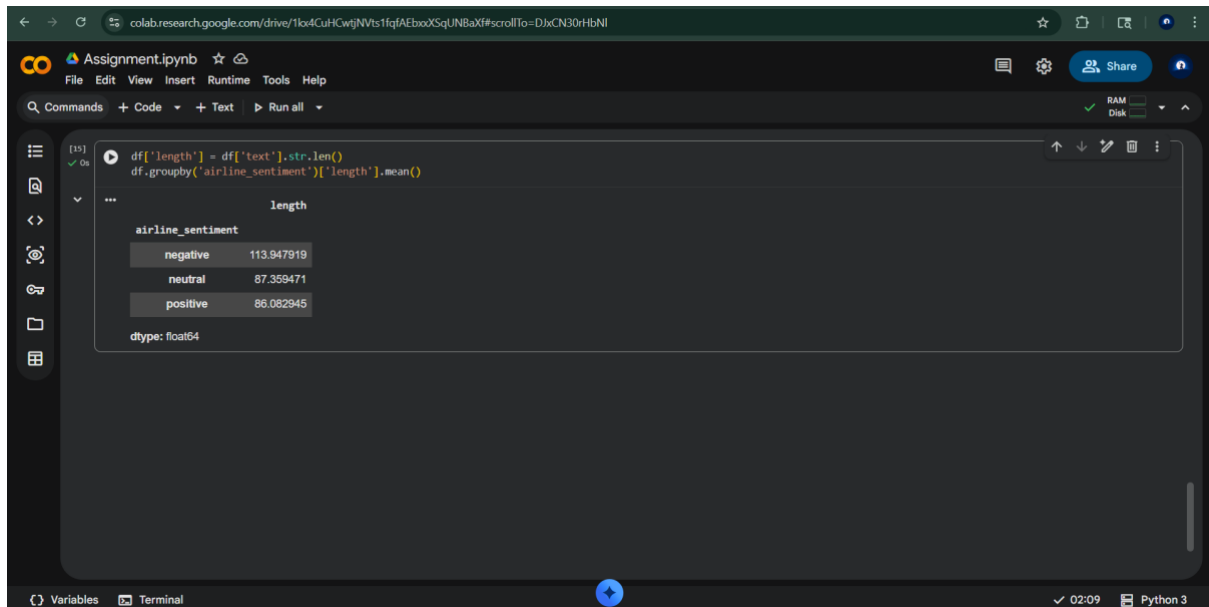
The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the following Python code:

```
df.loc[df['text'].str.len().idxmax()]
```

The output is a DataFrame with the following data:

| tweet_id           | airline_sentiment | airline_sentiment_confidence | negativereason | negativereason_confidence | airline    | airline_sentiment_gold | name    | negativereason_gold | retweet_count | text                                              | tweet_coord | tweet_created             | tweet_location                | user_timezone              | date       |
|--------------------|-------------------|------------------------------|----------------|---------------------------|------------|------------------------|---------|---------------------|---------------|---------------------------------------------------|-------------|---------------------------|-------------------------------|----------------------------|------------|
| 569328854948573187 | positive          | 0.6925                       | NaN            | 0.0                       | US Airways | NaN                    | zfelice | NaN                 | 0             | @US Airways Eyyyy! Cancelled Flightsations, Fl... | NaN         | 2015-02-21 18:52:31 -0800 | This Ain't Chicago, Tennessee | Central Time (US & Canada) | 2015-02-21 |

10. **Problem:** Compare tweet length across airline\_sentiment.



A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell [15] contains the following Python code:

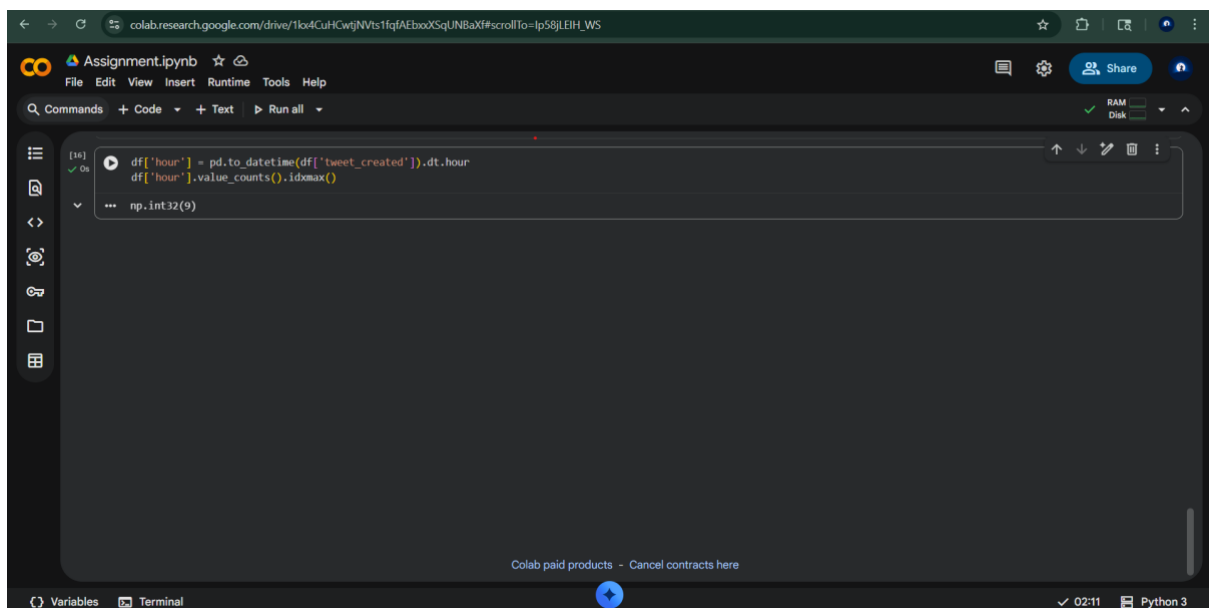
```
df['length'] = df['text'].str.len()
df.groupby('airline_sentiment')['length'].mean()
```

The output is a DataFrame with the following data:

| airline_sentiment | length     |
|-------------------|------------|
| negative          | 113.947919 |
| neutral           | 87.359471  |
| positive          | 86.082945  |

The output also shows the dtype as float64. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for commands, code, text, and running all cells, and a status bar at the bottom showing "Variables", "Terminal", "02:09", and "Python 3".

11. **Problem:** Find busiest hour using tweet\_created.

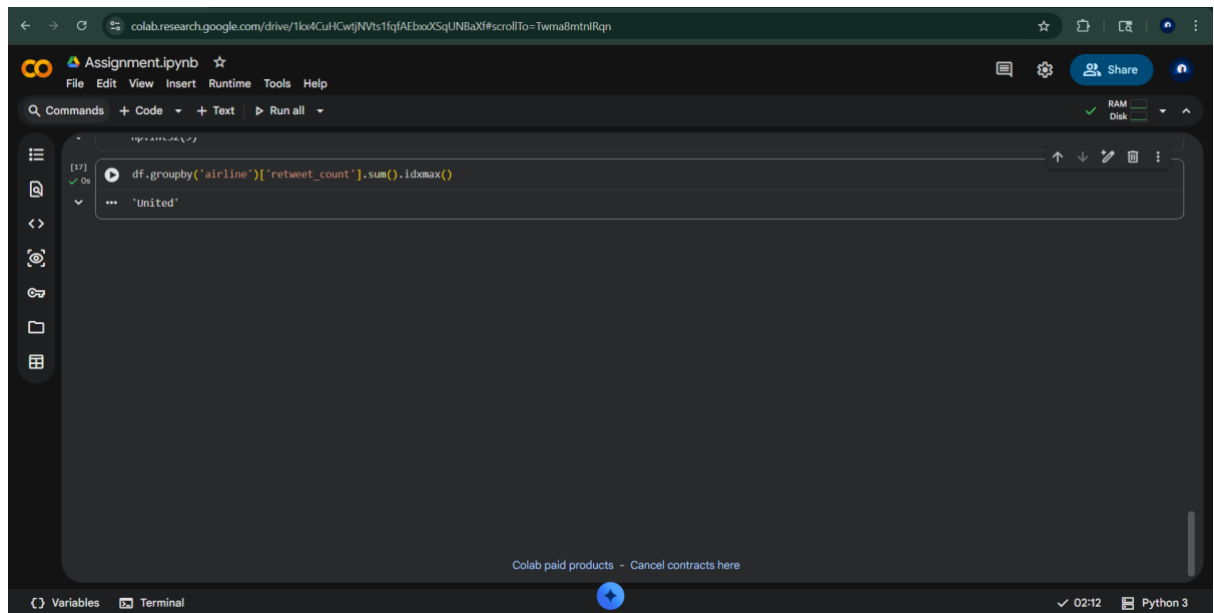


A screenshot of a Google Colab notebook titled "Assignment.ipynb". The code cell [16] contains the following Python code:

```
df['hour'] = pd.to_datetime(df['tweet_created']).dt.hour
df['hour'].value_counts().idxmax()
np.int32(9)
```

The notebook interface is identical to the previous screenshot, showing the same menu bar, toolbar, and status bar. The code cell is currently selected, and the output area is empty.

12. **Problem:** Find airline with highest total retweet\_count.

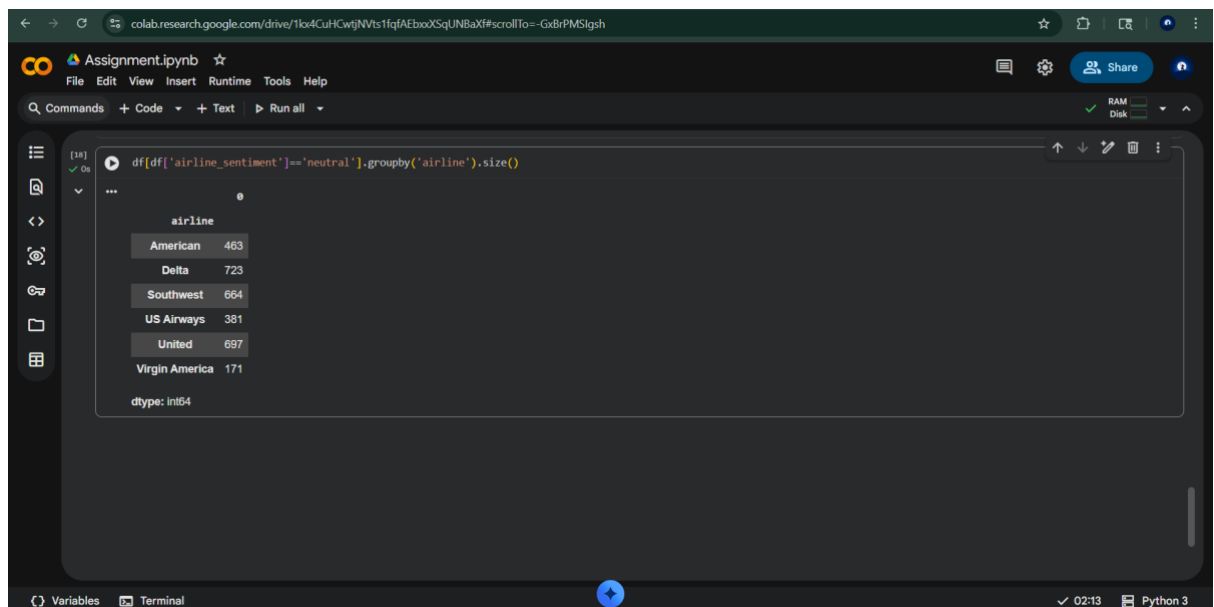


The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the following Python code:

```
[17]: df.groupby('airline')['retweet_count'].sum().idxmax()  
... 'United'
```

The output of the code is not visible in the image. The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Variables", "Terminal", and "Python 3".

13. **Problem:** Count neutral tweets airline-wise.

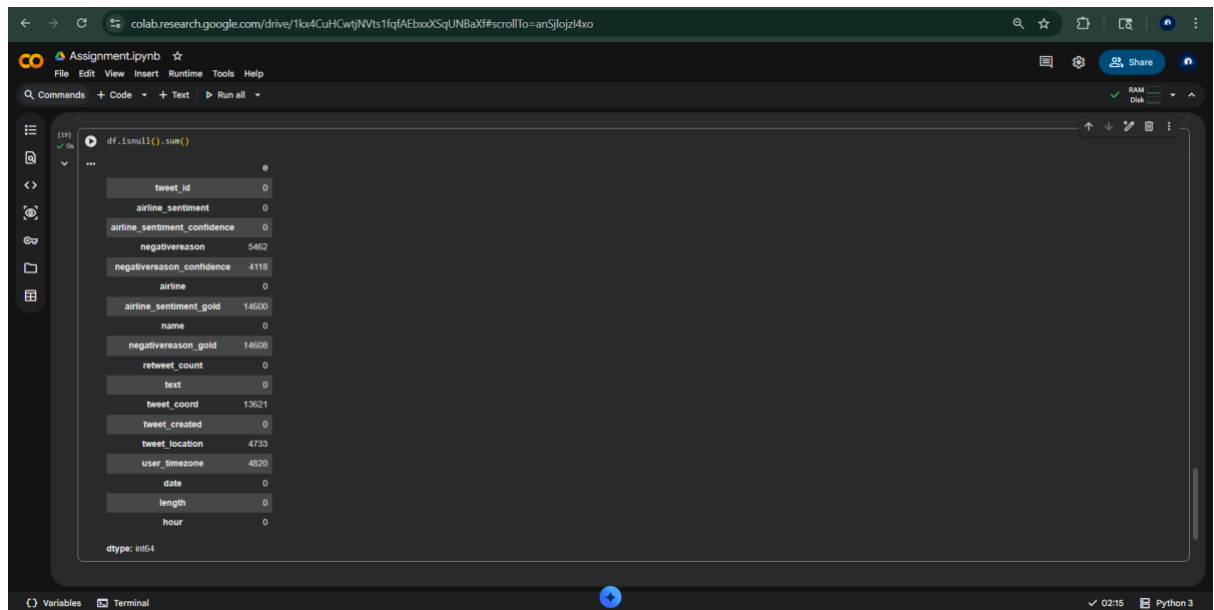


The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the following Python code:

```
[18]: df[df['airline_sentiment']=='neutral'].groupby('airline').size()  
...  
airline  
American    463  
Delta       723  
Southwest   664  
US Airways  381  
United      697  
Virgin America 171  
dtype: int64
```

The output of the code is a pandas Series showing the count of neutral tweets for each airline. The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Variables", "Terminal", and "Python 3".

14. **Problem:** Detect null values in all columns.

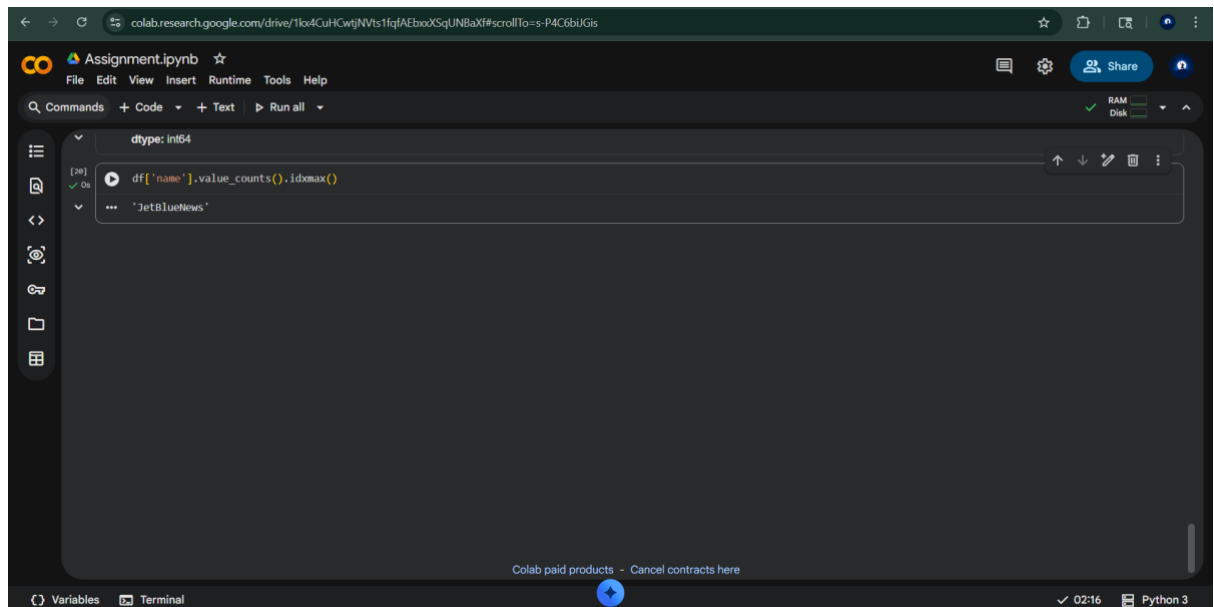


The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the command `df.isnull().sum()`, which has been executed. The output is a pandas Series showing the count of null values for each column in the DataFrame. The columns and their respective null counts are:

| Column                       | Count |
|------------------------------|-------|
| tweet_id                     | 0     |
| airline_sentiment            | 0     |
| airline_sentiment_confidence | 0     |
| negativereason               | 5462  |
| negativereason_confidence    | 4118  |
| airline                      | 0     |
| airline_sentiment_gold       | 14600 |
| name                         | 0     |
| negativereason_gold          | 14608 |
| retweet_count                | 0     |
| text                         | 0     |
| tweet_coord                  | 13621 |
| tweet_created                | 0     |
| tweet_location               | 4733  |
| user_timezone                | 4820  |
| date                         | 0     |
| length                       | 0     |
| hour                         | 0     |

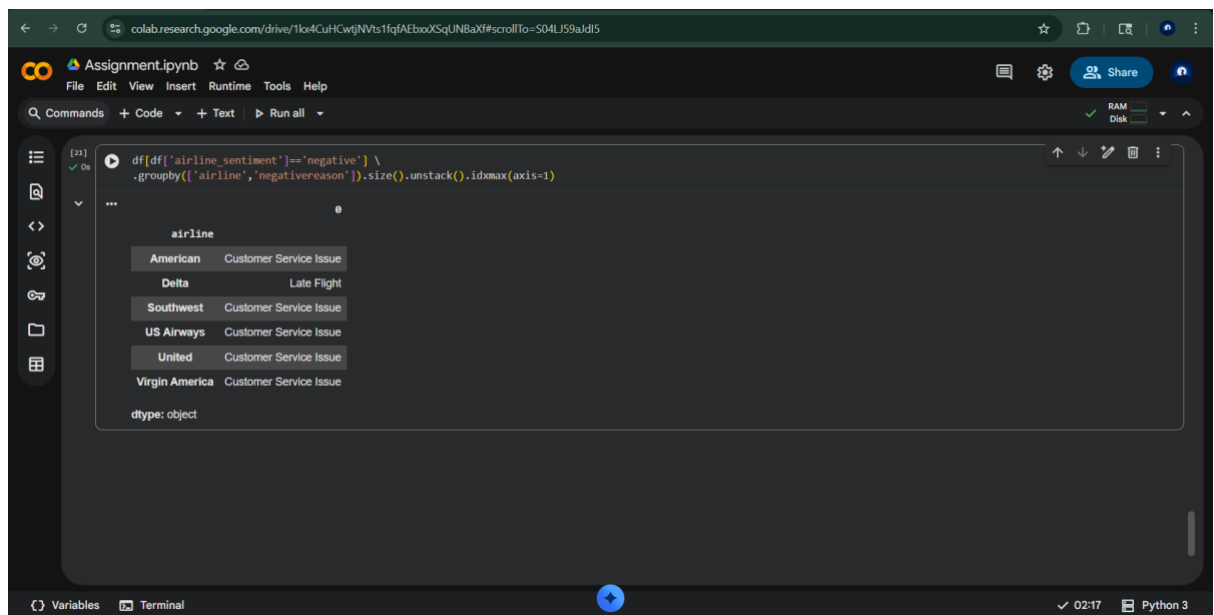
The output also indicates the data type: `dtype: int64`.

15. **Problem:** Find user (name) with highest tweet count.



The screenshot shows a Google Colab notebook titled "Assignment.ipynb". The code cell contains the command `df['name'].value_counts().idxmax()`, which has been executed. The output is the name of the user with the highest tweet count: `'JetBlueNews'`.

16. **Problem:** Find most common negativereason for each airline.



The screenshot shows a Google Colab notebook with the following code and output:

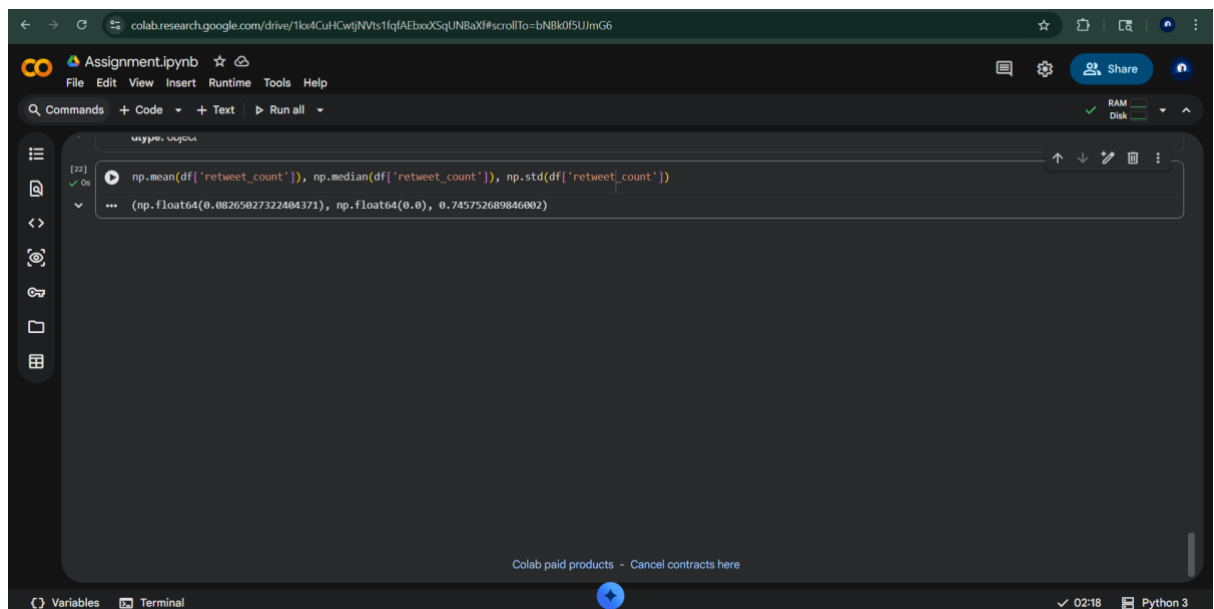
```
[21]: df[df['airline_sentiment']=='negative'] \
      .groupby(['airline', 'negativereason']).size().unstack().idxmax(axis=1)
```

The output is a DataFrame with the following structure:

| airline        |                        |
|----------------|------------------------|
| American       | Customer Service Issue |
| Delta          | Late Flight            |
| Southwest      | Customer Service Issue |
| US Airways     | Customer Service Issue |
| United         | Customer Service Issue |
| Virgin America | Customer Service Issue |

dtype: object

17. **Problem:** Calculate mean, median, std of retweet\_count.



The screenshot shows a Google Colab notebook with the following code and output:

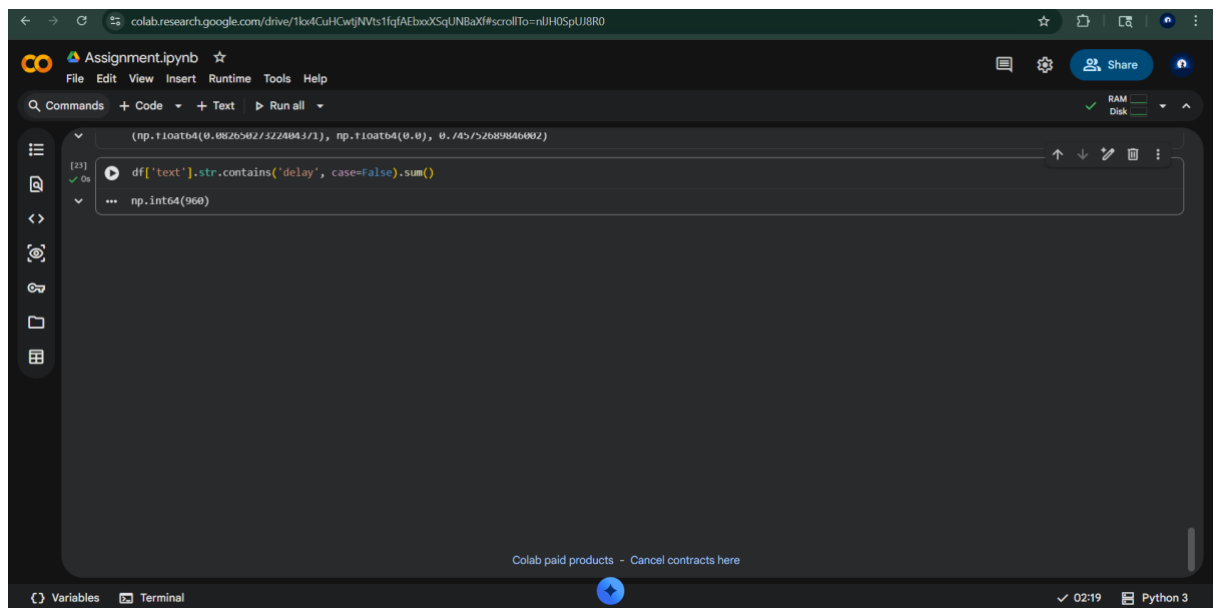
```
[22]: np.mean(df['retweet_count']), np.median(df['retweet_count']), np.std(df['retweet_count'])
```

The output is a tuple of three float64 values:

```
((np.float64(0.08265027322404371), np.float64(0.0), 0.745752689846002))
```



18. **Problem:** Count tweets mentioning "delay" in text.

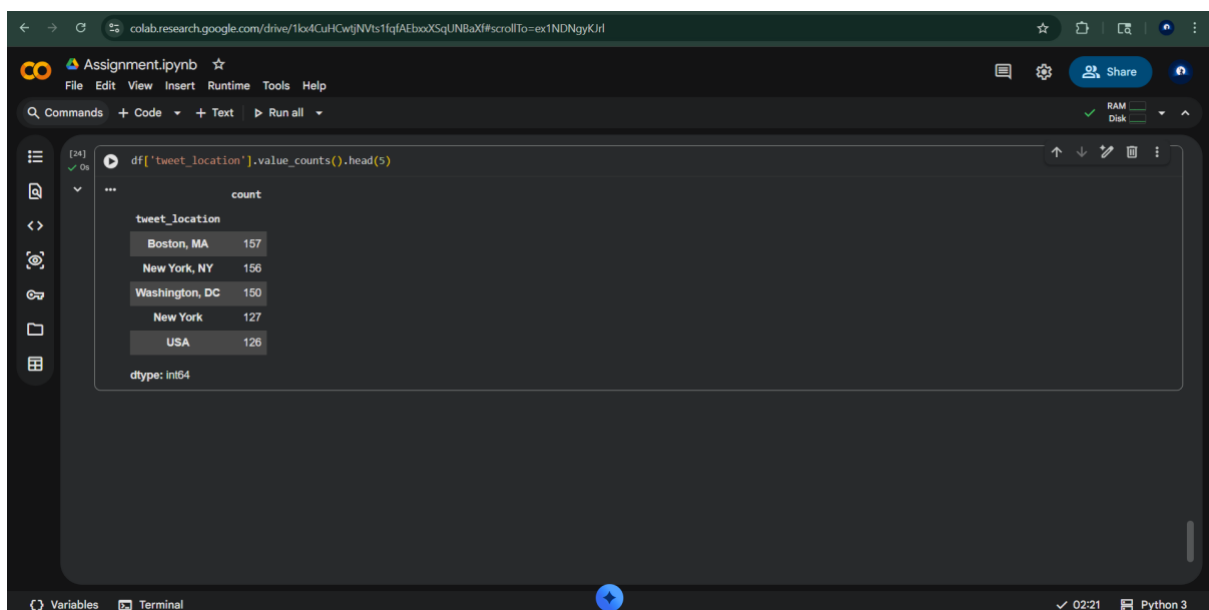


A Google Colab notebook titled "Assignment.ipynb" is shown. The code cell contains the following Python code:

```
(np.float64(0.082650273224043/1), np.float64(0.0), 0./45/52689846002)  
[23]: df['text'].str.contains('delay', case=False).sum()  
      np.int64(960)
```

The output of the code is displayed as a text cell with the value 960. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Colab paid products - Cancel contracts here", "Variables", "Terminal", and "Python 3".

19. **Problem:** Find most common tweet\_location.



A Google Colab notebook titled "Assignment.ipynb" is shown. The code cell contains the following Python code:

```
[24]: df['tweet_location'].value_counts().head(5)
```

The output of the code is displayed as a text cell showing the top 5 most common tweet locations and their counts:

| tweet_location | count |
|----------------|-------|
| Boston, MA     | 157   |
| New York, NY   | 156   |
| Washington, DC | 150   |
| New York       | 127   |
| USA            | 126   |

The output also shows the data type as int64. The notebook interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing "Variables", "Terminal", and "Python 3".

## 20. Problem: Identify outliers in retweet\_count using IQR.

colab.research.google.com/drive/1ko4CuHCWjNvts1fqAEboXsqUNBaXf#scrollTo=ex1NDNgYKrl

Assignment.ipynb

```
(In) [1]: Q1 = np.percentile(df['retweet_count'], 25)
Q3 = np.percentile(df['retweet_count'], 75)
IQR = Q3 - Q1
df[df['retweet_count'] > Q3 + 1.5 * IQR]
```

|       | tweet_id           | airline_sentiment | airline_sentiment_confidence | negativerreason         | negativerreason_confidence | airline        | airline_sentiment_gold | name          | negativerreason_gold | retweet_count | text                                               | tweet_coord                  | tweet_created             | tweet_location        | user_timezone               |
|-------|--------------------|-------------------|------------------------------|-------------------------|----------------------------|----------------|------------------------|---------------|----------------------|---------------|----------------------------------------------------|------------------------------|---------------------------|-----------------------|-----------------------------|
| 136   | 569625738231948800 | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | ktaptr        | NaN                  | 1             | @VirginAmerica You ROCK for making it so I can...  | NaN                          | 2015-02-22 14:32:14 -0800 | NaN                   | Mountain Time (US & Canada) |
| 147   | 569599867716132864 | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | jessicajaymes | NaN                  | 2             | Always have it together! You're welcome! RT ...    | (33.94652852, -118.40796257) | 2015-02-22 12:49:25 -0800 | hollywood, california | Pacific Time (US & Canada)  |
| 264   | 568829979981033472 | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | dchuang       | NaN                  | 1             | @VirginAmerica Thanks for gate checking my bag.    | (32.8454782, -96.8594585)    | 2015-02-29 16:27:32 -0800 | Orange County, CA     | Pacific Time (US & Canada)  |
| 304   | 568812048747201537 | negative          | 1.0                          | Lost Luggage            | 1.0000                     | Virgin America | NaN                    | sneilbell     | NaN                  | 1             | @VirginAmerica lost my luggage 4 days ago on f...  | NaN                          | 2015-02-29 08:36:54 -0800 | NaN                   | Quito                       |
| 318   | 568735060104511488 | neutral           | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | gemmabow19    | NaN                  | 2             | @VirginAmerica Can I help this & @fryallevan_f...  | NaN                          | 2015-02-29 03:32:59 -0800 | NaN                   | NaN                         |
| 14426 | 56861780885506448  | negative          | 1.0                          | Customer Service Issue  | 1.0000                     | American       | NaN                    | qpcqai        | NaN                  | 1             | @AmericanAir no one received text alerts, auto...  | NaN                          | 2015-02-22 14:00:43 -0800 | Chicago               | Central Time (US & Canada)  |
| 14524 | 569604328153063904 | negative          | 1.0                          | Late Flight             | 0.6849                     | American       | NaN                    | farazq        | NaN                  | 1             | @AmericanAir can you do anything to get #AA953...  | (40.88718573, -73.95477298)  | 2015-02-22 13:07:09 -0800 | New York, NY          | Quito                       |
| 14532 | 56960358927246336  | negative          | 1.0                          | Flight Booking Problems | 0.6964                     | American       | NaN                    | JasonShaw2    | NaN                  | 2             | @AmericanAir missing a full day of work thank...   | NaN                          | 2015-02-22 13:02:29 -0800 | Belleville            | Eastern Time (US & Canada)  |
| 14547 | 569601094238825920 | positive          | 1.0                          | NaN                     | NaN                        | American       | NaN                    | EricRoberts   | NaN                  | 1             | Eliza Kamp, I checked on u @AmericanAir with @...  | NaN                          | 2015-02-22 12:54:18 -0800 | Los Angeles, CA       | Pacific Time (US & Canada)  |
| 14619 | 56959888395708416  | positive          | 1.0                          | NaN                     | NaN                        | American       | NaN                    | howiemandel   | NaN                  | 3             | @AmericanAir I love your company and your staff... | NaN                          | 2015-02-22 12:14:08 -0800 | NaN                   | Pacific Time (US & Canada)  |

57 rows x 16 columns

colab.research.google.com/drive/1ko4CuHCWjNvts1fqAEboXsqUNBaXf#scrollTo=ex1NDNgYKrl

Assignment.ipynb

```
(In) [1]: Q1 = np.percentile(df['retweet_count'], 25)
Q3 = np.percentile(df['retweet_count'], 75)
IQR = Q3 - Q1
df[df['retweet_count'] > Q3 + 1.5 * IQR]
```

|  | airline_sentiment | airline_sentiment_confidence | negativerreason         | negativerreason_confidence | airline        | airline_sentiment_gold | name          | negativerreason_gold | retweet_count | text                                               | tweet_coord                  | tweet_created             | tweet_location        | user_timezone               | length | hour |
|--|-------------------|------------------------------|-------------------------|----------------------------|----------------|------------------------|---------------|----------------------|---------------|----------------------------------------------------|------------------------------|---------------------------|-----------------------|-----------------------------|--------|------|
|  | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | ktaptr        | NaN                  | 1             | @VirginAmerica You ROCK for making it so I can...  | NaN                          | 2015-02-22 14:32:14 -0800 | NaN                   | Mountain Time (US & Canada) | 106    | 14   |
|  | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | jessicajaymes | NaN                  | 2             | Always have it together! You're welcome! RT ...    | (33.94652852, -118.40796257) | 2015-02-22 12:49:25 -0800 | hollywood, california | Pacific Time (US & Canada)  | 95     | 12   |
|  | positive          | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | dchuang       | NaN                  | 1             | @VirginAmerica Thanks for gate checking my bag.    | (32.8454782, -96.8594585)    | 2015-02-29 16:27:32 -0800 | Orange County, CA     | Pacific Time (US & Canada)  | 122    | 16   |
|  | negative          | 1.0                          | Lost Luggage            | 1.0000                     | Virgin America | NaN                    | sneilbell     | NaN                  | 1             | @VirginAmerica lost my luggage 4 days ago on f...  | NaN                          | 2015-02-29 08:36:54 -0800 | NaN                   | Quito                       | 144    | 8    |
|  | neutral           | 1.0                          | NaN                     | NaN                        | Virgin America | NaN                    | gemmabow19    | NaN                  | 2             | @VirginAmerica Can I help this & @fryallevan_f...  | NaN                          | 2015-02-29 03:32:59 -0800 | NaN                   | NaN                         | 138    | 3    |
|  | negative          | 1.0                          | Customer Service Issue  | 1.0000                     | American       | NaN                    | qpcqai        | NaN                  | 1             | @AmericanAir no one received text alerts, auto...  | NaN                          | 2015-02-22 14:00:43 -0800 | Chicago               | Central Time (US & Canada)  | 126    | 14   |
|  | negative          | 1.0                          | Late Flight             | 0.6849                     | American       | NaN                    | farazq        | NaN                  | 1             | @AmericanAir can you do anything to get #AA953...  | (40.88718573, -73.95477298)  | 2015-02-22 13:07:09 -0800 | New York, NY          | Quito                       | 129    | 13   |
|  | negative          | 1.0                          | Flight Booking Problems | 0.6964                     | American       | NaN                    | JasonShaw2    | NaN                  | 2             | @AmericanAir missing a full day of work thank...   | NaN                          | 2015-02-22 13:02:29 -0800 | Belleville            | Eastern Time (US & Canada)  | 52     | 13   |
|  | positive          | 1.0                          | NaN                     | NaN                        | American       | NaN                    | EricRoberts   | NaN                  | 1             | Eliza Kamp, I checked on u @AmericanAir with @...  | NaN                          | 2015-02-22 12:54:18 -0800 | Los Angeles, CA       | Pacific Time (US & Canada)  | 113    | 12   |
|  | positive          | 1.0                          | NaN                     | NaN                        | American       | NaN                    | howiemandel   | NaN                  | 3             | @AmericanAir I love your company and your staff... | NaN                          | 2015-02-22 12:14:08 -0800 | NaN                   | Pacific Time (US & Canada)  | 113    | 12   |